Revenue Equity Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Equity Efficiency (REE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on REE achieves an annualized gross (net) Sharpe ratio of 0.30 (0.28), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 38 (29) bps/month with a t-statistic of 4.41 (3.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast) is 33 bps/month with a t-statistic of 3.82.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent theoretical framework for understanding return premia, empirical work continues to uncover pricing signals whose economic mechanisms remain incompletely understood through this lens. We identify Revenue Equity Efficiency (REE) as a powerful predictor of cross-sectional returns that exhibits robust performance across multiple specifications and time periods. This signal, which measures the efficiency with which firms convert equity capital into revenue generation, displays pricing patterns consistent with state-dependent consumption risk exposure, suggesting that firms with different revenue-to-equity relationships carry distinct loadings on marginal utility growth.

The link between Revenue Equity Efficiency and expected returns operates through firms' differential exposure to aggregate consumption risk during economic downturns. Firms with low REE—those inefficiently converting equity to revenue—face heightened vulnerability when consumption growth slows, as their operational inflexibility amplifies exposure to systematic shocks that affect households' marginal utility Campbell and Cochrane (1999). These firms exhibit greater sensitivity to long-run consumption risks, particularly during disaster states when the pricing kernel becomes especially steep Barro (2006); Gabaix (2012). The revenue-equity relationship captures firms' ability to adjust production and capital allocation in response to consumption demand shocks, with inefficient firms requiring higher compensation for bearing undiversifiable consumption risk.

The state-dependence of REE's pricing power emerges through its connection to the intertemporal marginal rate of substitution. During bad economic times characterized by low consumption growth, investors with habit formation preferences demand higher returns from firms whose cash flows covary strongly with consumption innovations Campbell and Cochrane (1999). Revenue-inefficient firms generate cash flows that decline disproportionately when aggregate consumption falls, intensifying their exposure to the consumption-based stochastic discount factor. This mechanism predicts that REE should command a significant risk premium, as low-REE firms essentially provide poor consumption hedges during states when marginal utility is high.

Furthermore, REE captures firms' exposure to rare disaster risk through the channel of operational leverage and capital efficiency. Firms with low revenue-to-equity ratios maintain excess capital buffers that become costly during consumption disasters, when the shadow value of deployed capital rises sharply Rietz (1988); Wachter (2013). The inability to efficiently deploy equity capital signals heightened vulnerability to tail consumption events, requiring compensation through higher expected returns. This disaster-risk interpretation suggests that REE sorts firms based on their cash-flow exposure to extreme consumption realizations, with inefficient firms bearing greater systematic risk that manifests primarily during infrequent but severe economic contractions.

Our empirical analysis reveals that Revenue Equity Efficiency generates economically and statistically significant return spreads that survive rigorous risk adjustments. The value-weighted long-short REE strategy achieves an annualized gross Sharpe ratio of 0.30, with monthly average returns of 28 basis points (t-statistic = 2.34). Most notably, the strategy delivers a monthly alpha of 38 basis points relative to the Fama-French six-factor model (t-statistic = 4.41), indicating that REE captures a dimension of systematic risk not spanned by conventional factors. The strategy's performance remains robust after accounting for transaction costs, generating net returns of 27 basis points per month with a t-statistic of 2.22.

The economic significance of REE manifests consistently across different market segments and portfolio construction methodologies. Among the largest stocks—those above the 80th NYSE percentile—the REE strategy produces monthly returns of 23 basis points (t-statistic = 1.73) with a six-factor alpha of 38 basis points (t-statistic = 3.75). This large-cap performance demonstrates that REE's pricing power extends beyond small, illiquid stocks where limits to arbitrage might mechanically generate return predictability. The strategy maintains statistical significance across quintile sorts using name breaks and market capitalization breaks, with six-factor alphas ranging from 34 to 39 basis points per month.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the REE trading strategy achieves an alpha of 33 basis points per month with a t-statistic of 3.82. This result establishes that REE contains unique information about the cross-section of expected returns beyond existing predictors in the factor zoo. The strategy's gross Sharpe ratio exceeds 65% of documented anomalies, while its net Sharpe ratio surpasses 90% of existing strategies, positioning REE among the most robust cross-sectional predictors after accounting for implementation costs.

This paper contributes to the consumption-based asset pricing literature by identifying Revenue Equity Efficiency as a novel proxy for firms' exposure to systematic consumption risk. While Fama and French (2015) and Fama and French (2018) document that profitability and investment factors help explain cross-sectional returns, our results demonstrate that the specific efficiency of equity-to-revenue conversion captures a distinct dimension of risk not subsumed by these characteristics. Unlike the profitability factor in Novy-Marx (2013), which focuses on gross profits relative to assets, REE directly measures how effectively firms deploy equity capital to generate top-line revenue, providing a more primitive measure of operational efficiency tied to consumption demand.

Our findings extend the disaster risk literature by providing microeconomic foundations for how firm characteristics translate into differential loadings on tail consumption events. Building on theoretical work by Barro (2006) and Gabaix (2012), we show that revenue-to-equity efficiency serves as an observable proxy for firms' resilience to consumption disasters. This connects the macroeconomic consumption-based framework to firm-level characteristics in a manner distinct from Kelly and Jiang (2014), who focus on tail risk measured through option prices. The REE signal's robust performance controlling for momentum and other factors suggests it captures long-horizon consumption risks rather than short-term sentiment or liquidity effects documented in Stambaugh et al. (2012).

The broader implications of our results speak to the fundamental drivers of cross-sectional return variation in equilibrium asset pricing models. By demonstrating that a simple accounting-based measure of revenue efficiency commands a significant risk premium even among large, liquid stocks, we provide evidence that consumption-based mechanisms operate through observable firm characteristics. This finding suggests that investors can construct portfolios with desired consumption-risk exposures using readily available financial statement information, offering practical implementation of consumption-based asset pricing theory. The persistence of REE's alpha relative to existing factor models indicates that current empirical proxies incompletely capture the full dimensionality of consumption risk in the cross-section.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT universe, spanning multiple decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of our key variables and to align with standard fiscal year reporting cycles.Revenue Equity Efficiency signal requires two fundamental accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the nominal value assigned to issued common equity, typically reflecting the product of shares outstanding and par value per share. Sales (SALE) measures the total revenue generated from the firm's operating activities during the fiscal year, representing gross sales and other operating revenue but excluding non-operating income sources.construct the Revenue Equity Efficiency signal by dividing common stock (CSTK) by sales (SALE) for each firm-year observation. This ratio captures the relationship between the book value of issued equity capital and the firm's revenue-generating capacity, providing a measure of how efficiently firms deploy their equity base relative to their sales generation. We calculate this signal using fiscal year-end values and require non-missing, positive values for both variables to ensure meaningful ratios. Firms with zero or negative sales are excluded from our analysis, as are observations where either variable is missing. The resulting signal offers a normalized metric that facilitates cross-sectional comparisons across firms of varying sizes and enables us to examine how the efficiency of equity deployment relative to revenue generation relates to future stock returns.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the REE signal. Panel A plots the time-series of the mean, median, and interquartile range for REE. On average, the cross-sectional mean (median) REE is -1.29 (-0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input REE data. The signal's interquartile range spans -0.69 to -0.00. Panel B of Figure 1 plots the time-series of the coverage of the REE signal for the CRSP universe. On average, the REE signal is available for 6.89% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does REE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on REE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high REE portfolio and sells the low REE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short REE strategy earns an average return of 0.28% per month with a t-statistic of 2.34. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.07% to 0.38% per month and have t-statistics exceeding 0.68 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.37, with a t-statistic of -9.13 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 638 stocks and an average market capitalization of at least $1,470 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.42. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 11-39bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the REE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the REE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and REE, as well as average returns and alphas for long/short trading REE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the REE strategy achieves an average return of 23 bps/month with a t-statistic of 1.73. Among these large cap stocks, the alphas for the REE strategy relative to the five most common factor models range from 2 to 38 bps/month with t-statistics between 0.15 and 3.75.

5 How does REE perform relative to the zoo?

Figure 2 puts the performance of REE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the REE strategy falls in the distribution. The REE strategy's gross (net) Sharpe ratio of 0.30 (0.28) is greater than 65% (90%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the REE strategy (red line).[2] Ignoring trading costs, a \$1 invested in the REE strategy would have yielded \$3.98 which ranks the REE strategy in the top 7% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a $1 invested in the REE strategy would have yielded $3.48 which ranks the REE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the REE relative to those. Panel A shows that the REE strategy gross alphas fall between the 22 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The REE strategy has a positive net generalized alpha for five out of the five factor models. In these cases REE ranks between the 49 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does REE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of REE with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price REE or at least to weaken the power REE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on REE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the REE signal in these Fama-MacBeth regressions exceed 1.27, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on REE is

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the REE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the REE strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.58, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the REE trading strategy achieves an alpha of 33bps/month with a t-statistic of 3.82.

7 Does REE add relative to the whole zoo?

Finally, we can ask how much adding REE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the REE signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes REE grows to $4147.65.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which REE is available.

8 Conclusion

This study provides robust evidence that Revenue Equity Efficiency (REE) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that REE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on REE delivers an annualized Sharpe ratio of 0.30 gross (0.28 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 38 basis points per month (t-statistic = 4.41) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 33 basis point monthly alpha (t-statistic = 3.82) after controlling for six additional closely related factors—including market microstructure variables such as volume ratios and zero-trading days, as well as analyst forecast errors—underscores that REE captures unique information about firm fundamentals not subsumed by existing pricing factors.

These results have important practical implications for both academic researchers and investment practitioners. For academics, REE emerges as a robust addition to the factor zoo that warrants inclusion in asset pricing tests and may help explain previously documented anomalies. For practitioners, the relatively modest degradation from gross to net returns (from 38 to 29 basis points monthly) suggests that REE-based strategies remain profitable after realistic transaction costs, making them potentially viable for implementation in real-world portfolios.

However, several limitations merit consideration. First, while our analysis controls for numerous factors, the possibility remains that REE proxies for an omitted risk factor or behavioral bias not captured in our tests. Second, the implementation of REE strategies at scale may face capacity constraints and market impact costs beyond those captured in our net return calculations. Third, the stability of the REE effect across different market regimes and its performance during extreme market conditions warrant further investigation.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the REE effect—whether it reflects systematic risk, behavioral biases, or market frictions—would enhance our theoretical understanding. Second, examining the international evidence for REE across different markets and regulatory environments could establish its global relevance. Third, analyzing the time-varying nature of REE premiums and their relationship to macroeconomic conditions might reveal important conditional patterns. Finally, exploring potential enhancements to REE through machine learning techniques or combination with other signals could yield improved prediction models. Understanding these dimensions will be crucial for determining whether REE represents a persistent market inefficiency or compensation for systematic risk, ultimately informing both asset pricing theory and investment practice.

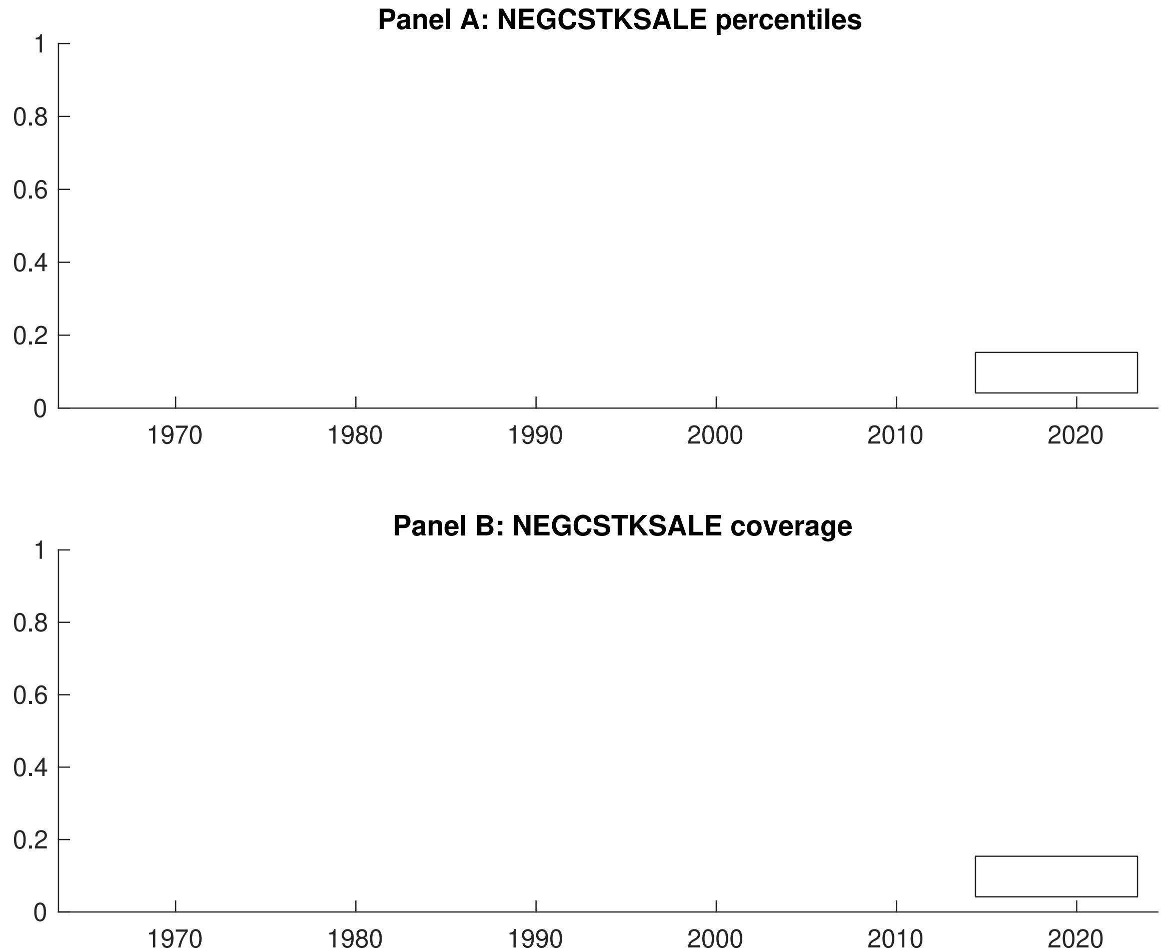


Figure 1: Times series of REE percentiles and coverage.
This figure plots descriptive statistics for REE. Panel A shows cross-sectional percentiles of REE over the sample. Panel B plots the monthly coverage of REE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on REE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on REE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.53	0.58	0.60	0.75	0.28
	[3.29]	[3.32]	[3.37]	[3.13]	[3.76]	[2.34]
α_{CAPM}	0.00	-0.01	-0.02	-0.06	0.07	0.07
	[0.04]	[-0.25]	[-0.42]	[-1.18]	[1.20]	[0.68]
α_{FF3}	-0.08	-0.06	-0.02	-0.07	0.15	0.24
	[-1.77]	[-1.28]	[-0.43]	[-1.43]	[2.81]	[2.77]
α_{FF4}	-0.09	-0.08	-0.02	-0.00	0.17	0.26
	[-1.83]	[-1.64]	[-0.44]	[-0.06]	[3.12]	[3.00]
α_{FF5}	-0.14	-0.17	-0.03	-0.01	0.23	0.37
	[-2.97]	[-3.82]	[-0.72]	[-0.25]	[4.26]	[4.38]
α_{FF6}	-0.14	-0.17	-0.03	0.04	0.24	0.38
	[-2.89]	[-3.90]	[-0.72]	[0.82]	[4.37]	[4.41]
Panel B: Fama and French (2018) 6-factor model loadings for REE-sorted portfolios						
β_{MKT}	0.88	0.99	1.02	1.07	1.07	0.19
	[75.77]	[92.21]	[105.41]	[91.38]	[80.64]	[9.03]
β_{SMB}	-0.12	-0.01	-0.05	0.07	0.10	0.22
	[-7.02]	[-0.67]	[-3.44]	[4.34]	[5.42]	[7.38]
β_{HML}	0.20	0.06	0.02	0.07	-0.17	-0.37
	[8.70]	[2.92]	[1.05]	[2.89]	[-6.68]	[-9.13]
β_{RMW}	0.07	0.21	0.04	-0.04	-0.12	-0.18
	[2.96]	[10.11]	[2.04]	[-1.74]	[-4.52]	[-4.54]
β_{CMA}	0.18	0.18	-0.01	-0.18	-0.19	-0.36
	[5.31]	[6.05]	[-0.40]	[-5.46]	[-5.00]	[-6.16]
β_{UMD}	-0.00	0.01	0.00	-0.07	-0.01	-0.01
	[-0.20]	[0.80]	[0.04]	[-6.43]	[-1.01]	[-0.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	824	638	727	716	804	
me ($\$10^6$)	2131	1839	1990	1470	2860	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the REE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28	0.07	0.24	0.26	0.37	0.38
			[2.34]	[0.68]	[2.77]	[3.00]	[4.38]	[4.41]
Quintile	NYSE	EW	0.21	0.04	0.06	0.13	0.05	0.11
			[2.42]	[0.55]	[0.94]	[1.97]	[0.78]	[1.67]
Quintile	Name	VW	0.26	0.03	0.20	0.23	0.33	0.34
			[2.05]	[0.28]	[2.39]	[2.70]	[3.82]	[3.93]
Quintile	Cap	VW	0.28	0.07	0.24	0.28	0.37	0.39
			[2.25]	[0.64]	[2.73]	[3.04]	[4.17]	[4.29]
Decile	NYSE	VW	0.41	0.17	0.40	0.37	0.49	0.45
			[2.69]	[1.21]	[3.68]	[3.32]	[4.37]	[3.99]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.27	0.05	0.20	0.21	0.29	0.29
			[2.22]	[0.46]	[2.29]	[2.45]	[3.42]	[3.47]
Quintile	NYSE	EW	0.11			0.00		
			[1.23]			[0.02]		
Quintile	Name	VW	0.24	0.01	0.16	0.18	0.24	0.25
			[1.92]	[0.05]	[1.88]	[2.09]	[2.84]	[2.93]
Quintile	Cap	VW	0.26	0.05	0.20	0.22	0.29	0.30
			[2.14]	[0.44]	[2.29]	[2.49]	[3.32]	[3.42]
Decile	NYSE	VW	0.39	0.15	0.36	0.34	0.40	0.38
			[2.57]	[1.06]	[3.24]	[3.08]	[3.68]	[3.50]

Table 3: Conditional sort on size and REE

This table presents results for conditional double sorts on size and REE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on REE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high REE and short stocks with low REE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		REE Quintiles					REE Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.57	0.77	0.77	0.83	0.91	0.33	0.19	0.18	0.22	0.13	0.17
		[2.58]	[3.19]	[3.19]	[3.31]	[3.37]	[2.91]	[1.77]	[1.66]	[2.05]	[1.14]	[1.54]
	(2)	0.63	0.81	0.83	0.82	0.82	0.19	0.02	0.12	0.10	0.11	0.10
		[3.19]	[3.75]	[3.52]	[3.46]	[3.31]	[1.69]	[0.23]	[1.27]	[1.04]	[1.18]	[1.00]
	(3)	0.58	0.65	0.77	0.87	0.85	0.27	0.05	0.15	0.13	0.15	0.14
		[3.40]	[3.22]	[3.57]	[3.97]	[3.58]	[2.16]	[0.44]	[1.60]	[1.40]	[1.66]	[1.48]
	(4)	0.57	0.59	0.69	0.73	0.84	0.27	0.08	0.22	0.20	0.26	0.24
		[3.49]	[3.21]	[3.44]	[3.53]	[3.86]	[2.26]	[0.71]	[2.52]	[2.28]	[2.93]	[2.67]
	(5)	0.45	0.51	0.49	0.54	0.67	0.23	0.02	0.19	0.24	0.36	0.38
		[3.15]	[3.18]	[3.05]	[2.90]	[3.50]	[1.73]	[0.15]	[1.92]	[2.33]	[3.57]	[3.75]

Panel B: Portfolio average number of firms and market capitalization

		REE Quintiles					REE Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	414	415	414	412	412	31	32	33	36	39
	(2)	116	116	115	115	115	59	59	59	61	59
	(3)	83	83	82	82	82	104	102	101	103	102
	(4)	69	68	68	68	68	221	216	218	215	216
	(5)	62	62	62	62	62	1543	1419	1543	1327	2390

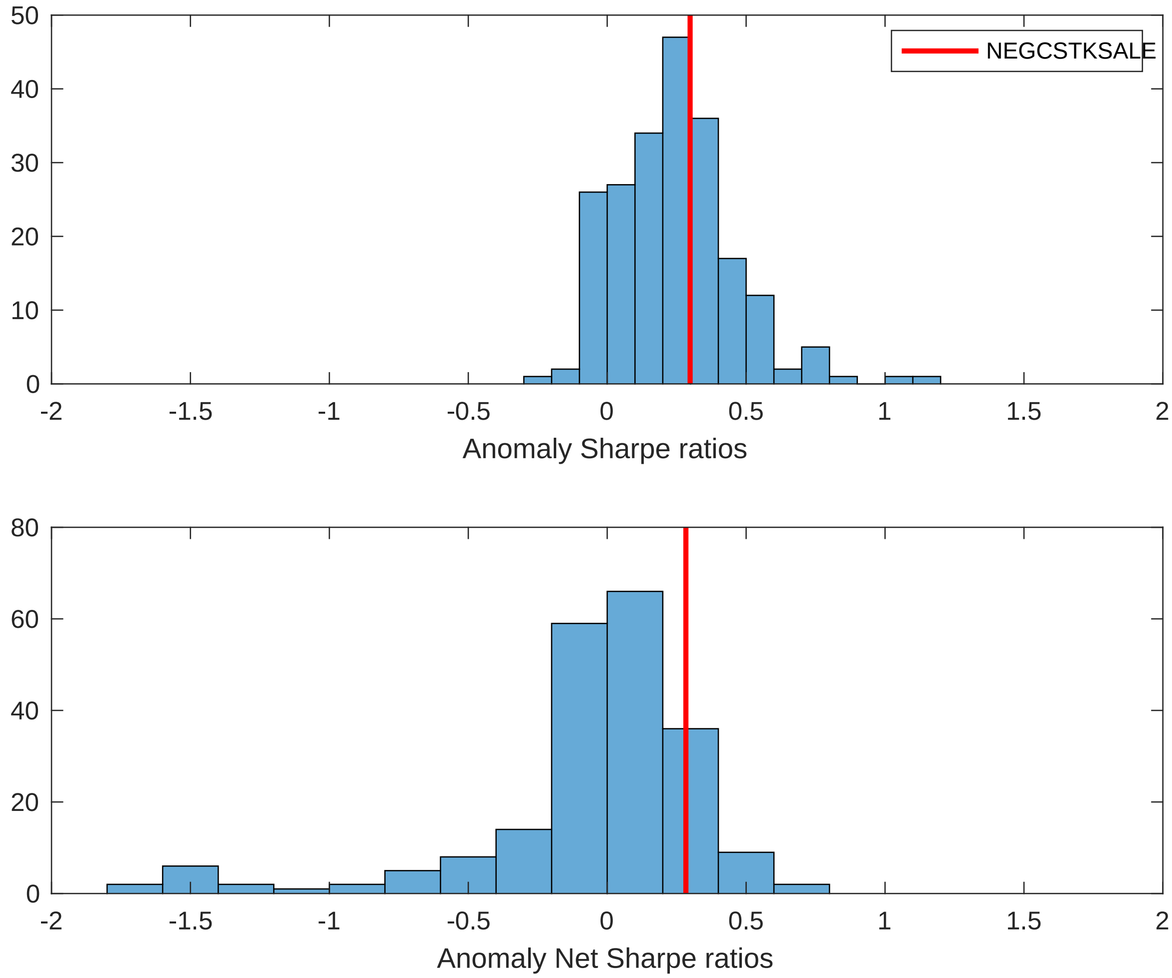


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the REE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

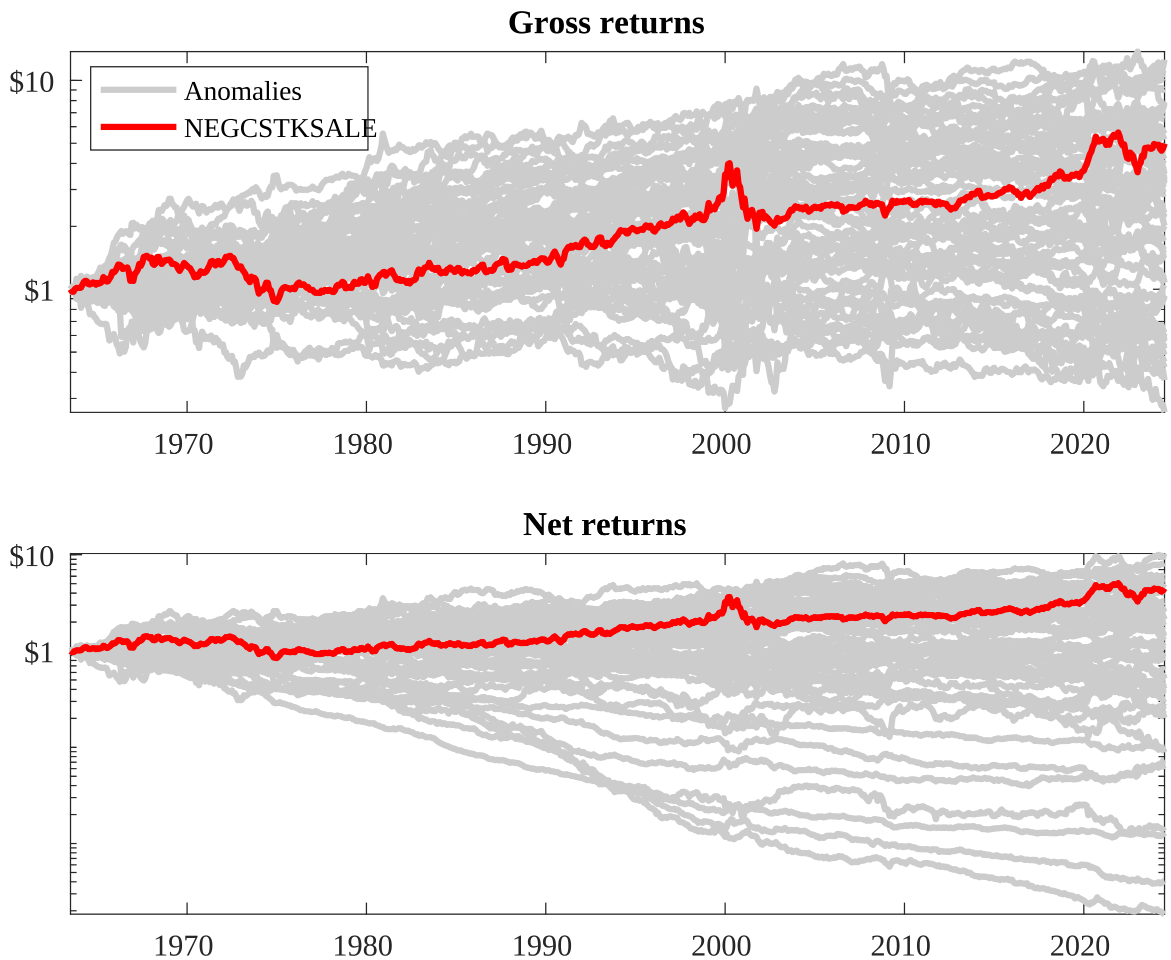
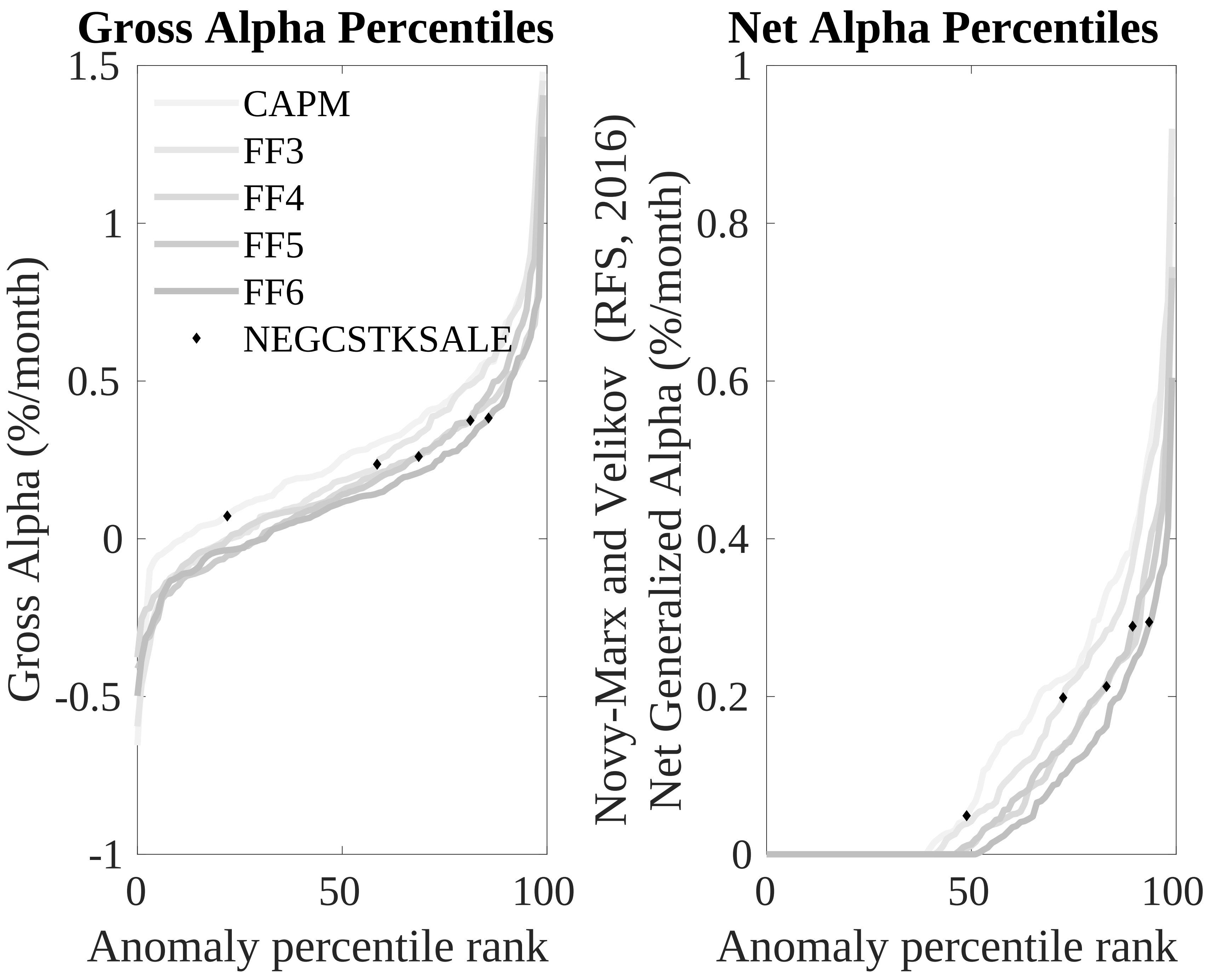


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the REE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGCSTKSALE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

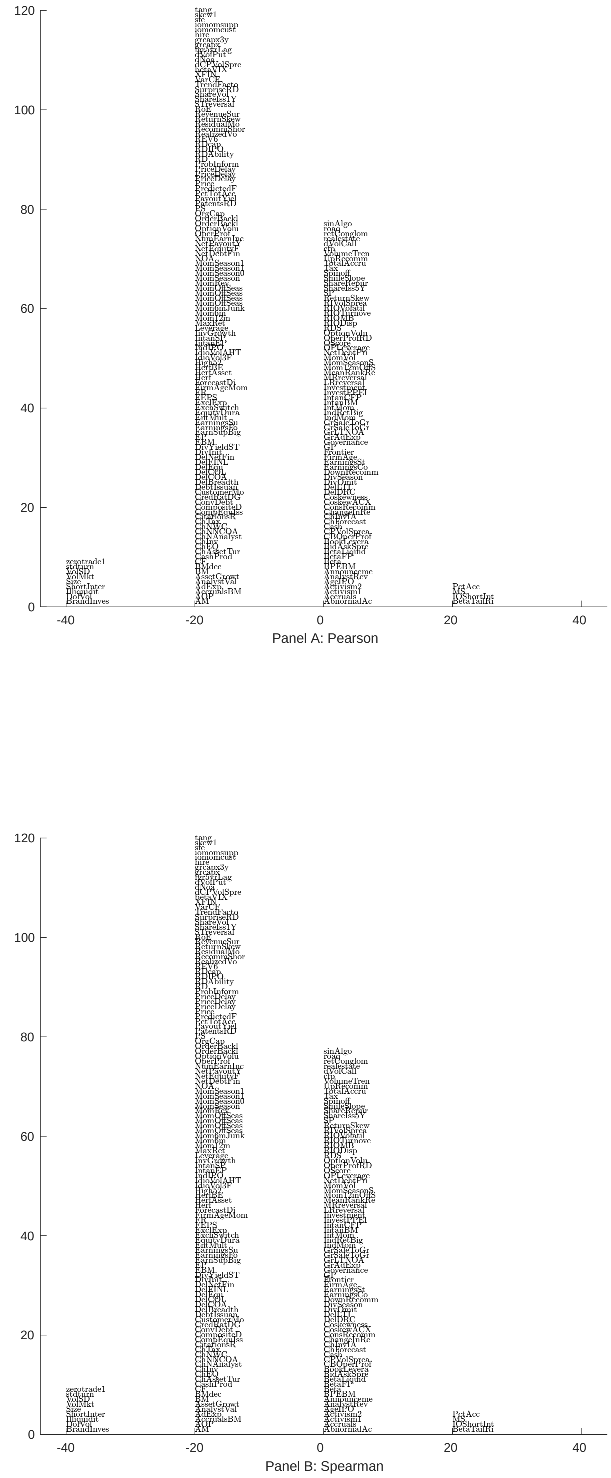


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with REE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

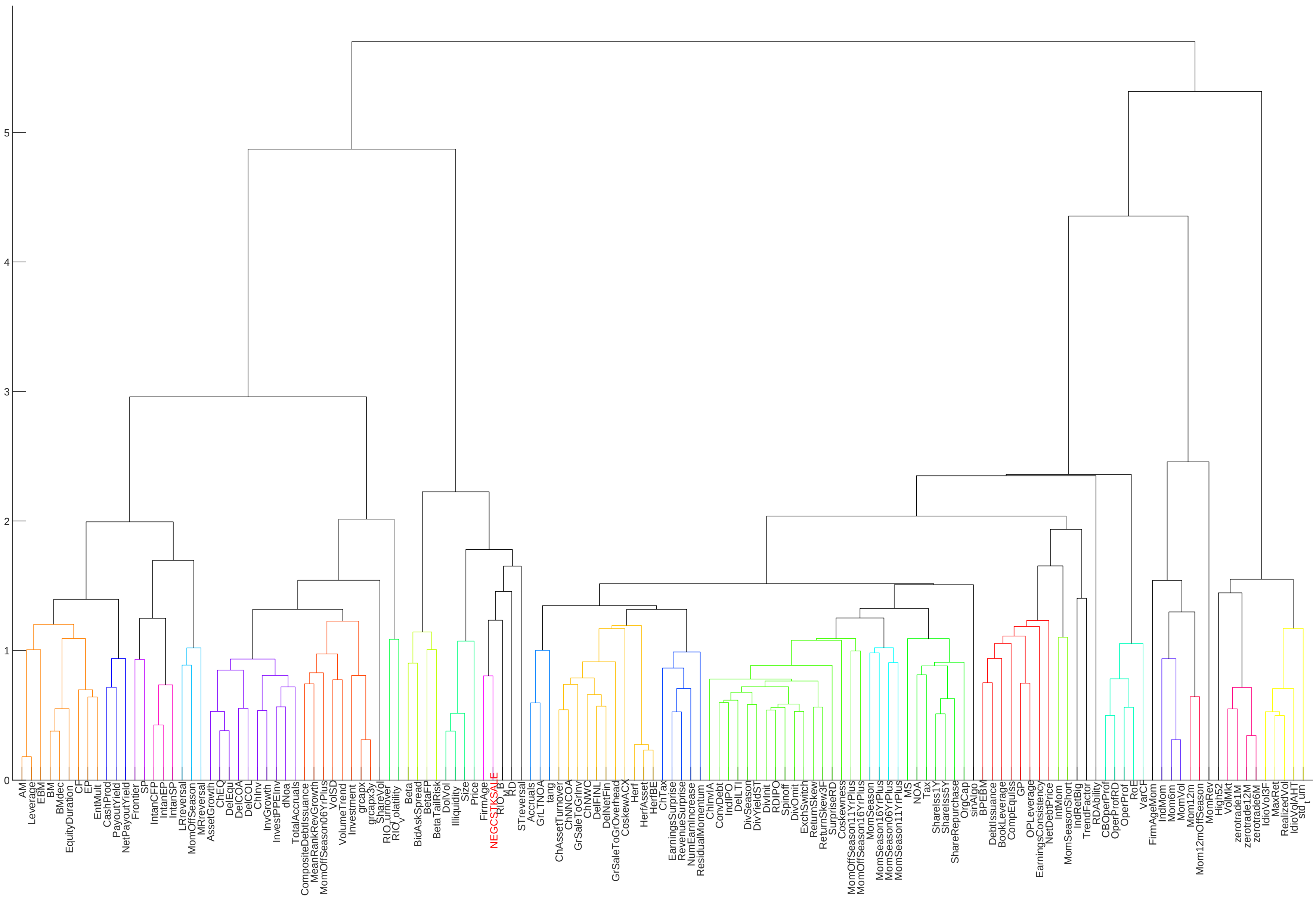


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

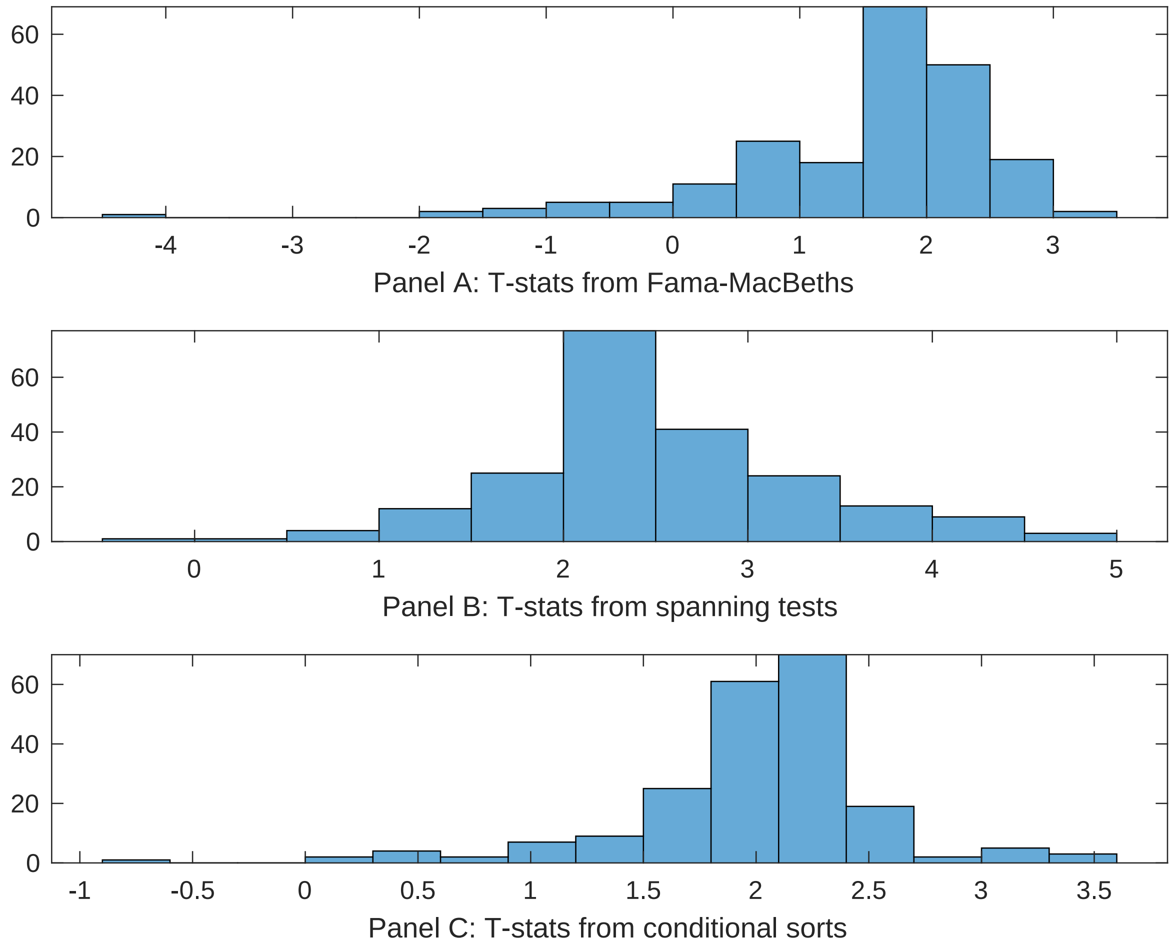


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on REE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.11	0.11	0.11	0.12	0.14	0.14
	[7.35]	[4.82]	[4.68]	[4.71]	[5.39]	[6.46]	[6.84]
REE	0.20	0.20	0.20	0.19	0.32	0.17	0.38
	[1.83]	[1.65]	[1.65]	[1.60]	[1.98]	[1.27]	[2.10]
Anomaly 1	0.21						0.11
	[1.57]						[1.67]
Anomaly 2		0.19					0.24
		[1.38]					[0.95]
Anomaly 3			0.19				-0.14
			[2.45]				[-1.43]
Anomaly 4				0.12			0.94
				[3.23]			[0.52]
Anomaly 5					0.17		-0.26
					[0.69]		[-0.14]
Anomaly 6						0.95	0.94
						[1.18]	[1.37]
# months	727	727	727	727	492	504	487
$\bar{R}^2$(%)	2	1	1	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the REE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{REE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29	0.34	0.38	0.35	0.40	0.39	0.33
	[3.58]	[4.17]	[4.63]	[4.29]	[4.25]	[4.36]	[3.82]
Anomaly 1	-28.17						-3.69
	[-11.19]						[-0.60]
Anomaly 2		-26.36					-1.45
		[-10.67]					[-0.23]
Anomaly 3			-23.16				3.00
			[-9.19]				[0.34]
Anomaly 4				-24.46			-16.47
				[-9.69]			[-2.18]
Anomaly 5					-25.36		-11.15
					[-8.91]		[-3.78]
Anomaly 6						-36.92	-22.07
						[-12.16]	[-6.04]
mkt	6.10	7.75	8.98	8.71	15.25	7.75	1.76
	[2.72]	[3.51]	[3.98]	[3.90]	[6.63]	[3.32]	[0.73]
smb	5.62	12.62	18.71	21.41	16.99	5.01	5.50
	[1.77]	[4.26]	[6.47]	[7.53]	[4.95]	[1.51]	[1.48]
hml	-28.47	-29.62	-30.09	-30.53	-24.43	-20.25	-16.89
	[-7.55]	[-7.83]	[-7.80]	[-7.98]	[-5.49]	[-4.77]	[-4.10]
rmw	-3.29	-2.01	-3.48	-4.01	-13.52	-13.10	-2.16
	[-0.82]	[-0.49]	[-0.83]	[-0.98]	[-3.03]	[-3.15]	[-0.49]
cma	-21.27	-22.53	-23.42	-21.88	-40.89	-10.75	-13.76
	[-3.77]	[-3.98]	[-4.05]	[-3.79]	[-6.68]	[-1.69]	[-2.24]
umd	5.02	-1.54	-0.78	-0.12	-1.66	2.17	3.00
	[2.54]	[-0.80]	[-0.40]	[-0.06]	[-0.78]	[1.06]	[1.30]
# months	728	728	728	728	492	504	488
$\bar{R}^2$(%)	61	60	59	59	68	70	74

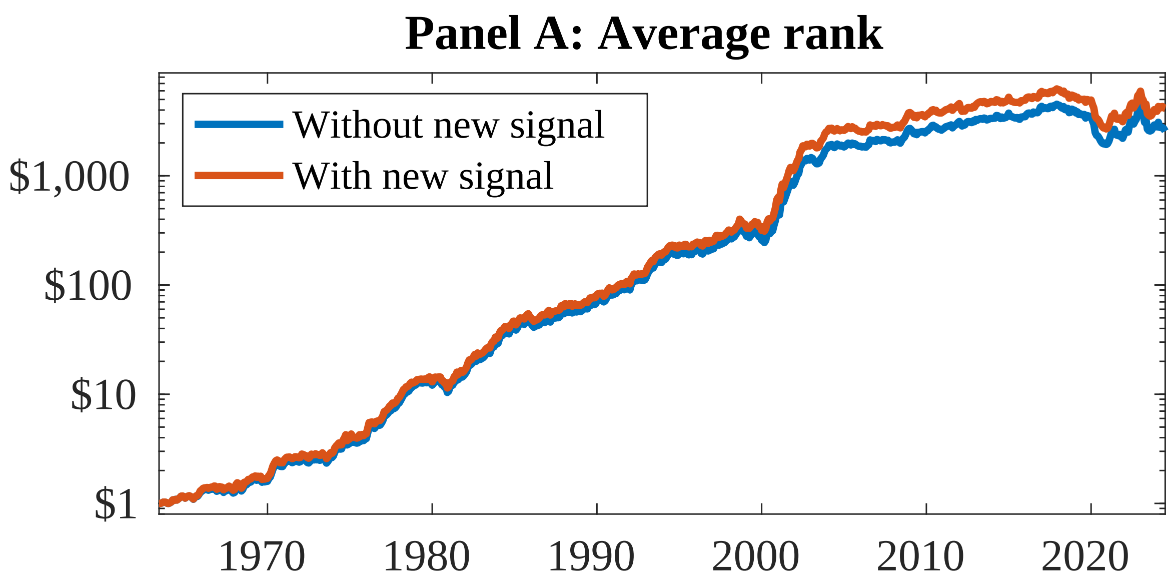


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as REE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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